QED radiative corrections for exclusive meson electroproduction $ep \to e'\,B\,h$ ($h = \rho^0, \phi, J/\psi$ or π^0, π^+, η) with the full set of structure functions $\sigma_T,\ \sigma_L,\ \sigma_{TT},\ \sigma_{LT},\ \sigma_{LT'}$

Formalism and description of the `vmrad` program

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1 Motivation

The program DIFFRAD [1] computes QED radiative corrections (RC) to exclusive vector–meson electroproduction

$$e(k_1) + p(p) \ \longrightarrow\ e'(k_2) + p'(p') + V(p_v), \qquad V = \rho^0,\ \phi,\ J/\psi, \tag{1}$$

using only the two structure functions σ_T and σ_L of the ϕ-integrated cross section. The RC, however, depends on the structure functions (SFs) used as input: the radiative "tail" entering a given observed bin is built from the cross section at *shifted* (true) kinematics, and the azimuthal modulations generated by $\sigma_{TT}, \sigma_{LT}, \sigma_{LT'}$ both receive and induce corrections that a two-SF calculation cannot describe. Here we write down the RC for the complete set of five SFs, including beam polarization, in the exact covariant approach of Bardin and Shumeiko [2] as implemented for exclusive pion electroproduction in EXCLURAD [3], adapted to vector–meson production. As in DIFFRAD/EXCLURAD, radiation from the lepton line, the lepton vertex correction and vacuum polarization are taken into account; radiation from the hadronic line and two–photon exchange are neglected.

2 Kinematics and conventions

Metric $(+,-,-,-)$; m is the lepton mass, M the proton mass, m_V the meson mass. Observed (reconstructed from e' and V) variables:

$$q = k_1 - k_2, \quad Q^2 = -q^2, \quad W^2 = (p+q)^2, \quad t = (q - p_v)^2, \quad \phi, \tag{2}$$

with ϕ the azimuth of the hadron plane about $\boldsymbol{q}$ measured from the lepton plane. All four–vectors are constructed in the CM frame of $(p+q)$: $\boldsymbol{q}$ along $+z$, leptons in the x–z plane with $k_{1x} > 0$, $y = z \times x$ (Trento–like). Invariants:

$$S = 2k_1{\cdot}p, \quad X = 2k_2{\cdot}p = S - S_x, \quad S_x = 2p{\cdot}q = W^2 + Q^2 - M^2, \tag{3}$$

$$\lambda_S = S^2 - 4m^2M^2, \qquad \lambda_q = S_x^2 + 4M^2Q^2, \qquad y = S_x/S. \tag{4}$$

The virtual–photon polarization parameter (massless–lepton form, used only for quoting the canonical decomposition; the code keeps m exactly)

$$\varepsilon = \frac{1 - y - \frac{1}{4}\gamma^2y^2}{1 - y + \frac{1}{2}y^2 + \frac{1}{4}\gamma^2y^2}, \qquad \gamma^2 = \frac{4M^2x_B^2}{Q^2}, \quad x_B = \frac{Q^2}{S_x}. \tag{5}$$

For the radiative process $ep \to e'p'V\gamma(k)$ the reconstructed missing mass moves away from the proton mass. We use the *inelasticity*

$$v \equiv \Lambda^2 - M^2, \qquad \Lambda \equiv p + q - p_v = p' + k, \tag{6}$$

so $v = M_X^2 - M^2$: an experimental missing–mass cut $M_X^2 < M_{X,\mathrm{cut}}^2$ is exactly a cut $v < v_{\mathrm{cut}}$. At the Born level $v = 0$. At fixed (Q^2, W^2, t) the meson CM energy is $E_v = (W^2 + m_V^2 - M^2 - v)/2W$ and

$$\cos\theta_v = \frac{t + Q^2 - m_V^2 + 2q_0 E_v}{2|\boldsymbol{q}||\boldsymbol{p}_v|}, \tag{7}$$

which yields the kinematic limit $v_{\max}^{\mathrm{kin}}(Q^2, W^2, t)$ from $|\cos\theta_v| \le 1$ (bounded globally by $(W - m_V)^2 - M^2$). The integration limit is $v_m = \min(v_{\max}^{\mathrm{kin}}, v_{\mathrm{cut}})$.

3 Born cross section and structure functions

The Born matrix element is $\mathcal{M}_0 = (e^2/Q^2)\, j_\mu^0 H^\mu$ with $j_\mu^0 = \bar{u}(k_2)\gamma_\mu u(k_1)$. With the exact three–body phase space one finds

$$\boxed{\frac{d\sigma_0}{dQ^2\, dW^2\, dt\, d\phi} = \frac{\alpha^2}{32\pi^2\, \lambda_S\, \lambda_q^{1/2}\, Q^4}\; L_{\mu\nu}^0 H^{\mu\nu}} \tag{8}$$

where $L_{\mu\nu}^0(h) = \sum_{\mathrm{spins}} j_\mu^0 j_\nu^{0*}$ for beam helicity $h = \pm 1$, and $H^{\mu\nu} = \overline{\sum} H^\mu H^{\nu *}$ is the hadronic tensor of $\gamma^* p \to V p'$.

Structure function definitions. Let ℓ^μ (longitudinal) and X^μ, Y^μ (transverse, X in the hadron plane, Y normal to it) be the photon polarization basis in the $\gamma^* p$ CM. The SFs are Hand–convention virtual–photoproduction cross sections,

$$\frac{d\sigma_{\lambda\lambda'}}{dt} = \frac{e^2\, \epsilon_\lambda^{\mu *} H_{\mu\nu} \epsilon_{\lambda'}^{\nu}}{16\pi\, (W^2 - M^2)^2}, \tag{9}$$

in terms of which the hadronic tensor reads

$$\boxed{H^{\mu\nu} = \frac{\kappa}{2}\Big[\sigma_T\big(X^\mu X^\nu + Y^\mu Y^\nu\big) + \sigma_L\, \ell^\mu \ell^\nu + \sigma_{TT}\big(X^\mu X^\nu - Y^\mu Y^\nu\big) - \sigma_{LT}\big(\ell^\mu X^\nu + X^\mu \ell^\nu\big) - i\,\sigma_{LT'}\big(\ell^\mu X^\nu - X^\mu \ell^\nu\big)\Big]} \tag{10}$$

with $\kappa = 8(W^2 - M^2)^2/\alpha$ (each $\sigma_i = d\sigma_i/dt$). Inserting (10) into (8) reproduces *identically* (this is verified to machine precision in the code, for both beam helicities — the "Born closure" test) the canonical decomposition

$$\frac{d\sigma_0}{dQ^2 dW^2 dt\, d\phi} = \frac{\Gamma_v}{2\pi}\Big[\sigma_T + \varepsilon\,\sigma_L + \varepsilon\cos 2\phi\,\sigma_{TT} + \sqrt{2\varepsilon(1+\varepsilon)}\cos\phi\,\sigma_{LT} + h\sqrt{2\varepsilon(1-\varepsilon)}\sin\phi\,\sigma_{LT'}\Big]. \tag{11}$$

If your SF fits use $\sqrt{\varepsilon(1+\varepsilon)}$ instead of $\sqrt{2\varepsilon(1+\varepsilon)}$, or opposite signs for σ_{TT}, σ_{LT}, rescale before plugging them in.

4 Radiative corrections: master formula

With the Bardin–Shumeiko covariant separation of the infrared divergence (no soft–photon cutoff parameter appears anywhere),

$$\boxed{\sigma_{\mathrm{obs}} = \sigma_0\, e^{\delta_{\mathrm{inf}}}\,(1 + \delta_{VR} + \delta_{\mathrm{vac}}) + \sigma_F} \tag{12}$$

(EXCLURAD Eq. (55) [3]); the exponentiation of δ_{inf} accounts for multiple soft photons.

4.1 Virtual and soft parts

Let $l_m = \ln(Q^2/m^2)$ and

$$S'_0 = 2k_1\cdot p'_B, \qquad X'_0 = 2k_2\cdot p'_B, \qquad p'_B = p + q - p_v\big|_{v=0} \ \text{(Born recoil proton)}. \tag{13}$$

Then

$$\delta_{\rm inf} = \frac{\alpha}{\pi}\,(l_m - 1)\,\ln\frac{v_m^2}{S'_0 X'_0}, \tag{14}$$

$$\delta_{VR} = \frac{\alpha}{\pi}\left(\frac{3}{2}l_m - 2 - \frac{1}{2}\ln^2\frac{X'_0}{S'_0} + \mathrm{Li}_2\Big[1 - \frac{Q^2M^2}{S'_0X'_0}\Big] - \frac{\pi^2}{6}\right). \tag{15}$$

Note that S'_0, X'_0 depend on the hadron kinematics (through p_v), so $\delta_{VR}, \delta_{\rm inf}$ vary with t and ϕ as well as with the missing–mass cut through v_m.

Vacuum polarization: $\delta_{\rm vac} = \delta_{\rm vac}^{\rm lept} + \delta_{\rm vac}^{\rm had}$ with the exact one–loop leptonic part

$$\delta_{\rm vac}^{\rm lept} = 2\sum_{\ell=e,\mu,\tau}\bar\Pi_\ell(Q^2), \qquad \bar\Pi_\ell = \frac{2\alpha}{\pi}\int_0^1 dx\,x(1-x)\ln\Big[1 + \frac{Q^2x(1-x)}{m_\ell^2}\Big], \tag{16}$$

and the standard hadronic parameterization $\delta_{\rm vac}^{\rm had} = -2\big(a + b\ln(1+cQ^2)\big)$ with the usual piecewise coefficients (as in the POLRAD/HAPRAD family).

4.2 Hard–photon (finite) part σ_F

For the radiative process the true virtual photon is $\tilde q = q - k$ and the hadronic vertex is evaluated at the *shifted* kinematics

$$\tilde Q^2 = Q^2 + 2k\cdot q, \qquad \tilde W^2 = W^2 - 2k\cdot(p+q), \qquad \tilde t = t - 2k\cdot(q - p_v). \tag{17}$$

Eliminating the unobserved proton with $\delta^{(4)}$ and its mass shell, the photon phase space reduces, at fixed observed (Q^2, W^2, t, ϕ), to an integral over v and the photon direction $\boldsymbol{n}$: with

$$z \equiv \frac{k\cdot\Lambda}{\omega} = \Lambda_0 - \boldsymbol{\Lambda}\cdot\boldsymbol{n}, \qquad \omega = \frac{v}{2z}, \tag{18}$$

the exact radiative cross section is

$$\frac{d\sigma_R}{dQ^2dW^2dt\,d\phi} = \frac{\alpha^3}{512\,\pi^4\,\lambda_S\,\lambda_q^{1/2}}\int_0^{v_m} dv\int d\Omega_k\;\frac{v}{z^2}\;\frac{L^R_{\mu\nu}\,\tilde H^{\mu\nu}}{\tilde Q^4}, \tag{19}$$

where $\tilde H^{\mu\nu}$ is the tensor (10) built from $\tilde q$, p, $p_v(v)$ with all five SFs evaluated at $(\tilde W^2, \tilde Q^2, \tilde t)$, and $L^R_{\mu\nu}(h)$ is the exact radiative leptonic tensor

$$L^R_{\mu\nu} = \sum_{\text{spins, pol}} j^R_\mu j^{R*}_\nu, \qquad j^R_\mu = \bar u(k_2)\Big[\gamma_\mu\frac{\not k_1 - \not k + m}{-\varkappa_1}\not\epsilon^* + \not\epsilon^*\frac{\not k_2 + \not k + m}{\varkappa_2}\gamma_\mu\Big]u(k_1,h), \tag{20}$$

$\varkappa_{1,2} = 2k\cdot k_{1,2}$. In the soft limit $L^R \to 4F_{\rm IR}L^0$ with the eikonal factor

$$F_{\rm IR} = \frac{Q^2 + 2m^2}{\varkappa_1\varkappa_2} - \frac{m^2}{\varkappa_1^2} - \frac{m^2}{\varkappa_2^2}. \tag{21}$$

The IR divergence of (19) at $v \to 0$ is removed covariantly by subtracting the eikonal times the *unshifted* Born integrand — whose analytic integral is precisely what produced (14)–(15) — leaving the finite

$$\boxed{\sigma_F = \frac{\alpha^3}{512\pi^4 \lambda_S \lambda_q^{1/2}} \int_0^{v_m} dv \int d\Omega_k \, \frac{v}{z^2} \left[\frac{L^R_{\mu\nu} \tilde{H}^{\mu\nu}}{\tilde{Q}^4} - 4F_{\rm IR} \, \frac{L^0_{\mu\nu} H^{0\,\mu\nu}}{Q^4} \right]} \tag{22}$$

(the analogue of EXCLURAD Eq. (38)). Every azimuthal interference structure of all five SFs at shifted kinematics is generated *exactly* by the tensor contraction $L^R : \tilde{H}$ — no leading–log or peaking approximation is made.

5 Numerical implementation (`vmrad`)

- L^0, L^R are evaluated *numerically* from helicity spinors (exact in m, exact beam polarization), instead of transcribing the very long analytic contractions of [3]. Correctness is enforced by the validation suite: Born closure against (11) ($\sim 3 \times 10^{-8}$, limited only by lepton–mass terms), current conservation $\tilde{q}_\mu L^{R\,\mu\nu} = 0$ ($\sim 10^{-9}$), and soft factorization $L^R \to 4F_{\rm IR} L^0$.
- The polarization basis (X, Y, ℓ) at shifted kinematics is built by boosting to the $(p + \tilde{q})$ CM; the SFs enter as scalars $\sigma_i(\tilde{W}^2, \tilde{Q}^2, \tilde{t})$ — no shifted azimuth is ever needed.
- The photon angular integral uses a two–channel decomposition with weights $w_c \propto \varkappa_c^{-2}$ centered on $\boldsymbol{k}_1$ and $\boldsymbol{k}_2$ and the substitution $x = \ln(a_c - \cos\theta_c)$, which flattens the collinear peaks (width m^2/E^2); the collinear core gets a dedicated Gauss–Legendre panel. The v integral is Gauss–Legendre in $\ln v$.
- IR bookkeeping is verified non–circularly: the angular integral of the subtraction term in (22) reproduces the analytic slope $d\,\delta_{\rm inf}/d\ln v^2 = (\alpha/\pi)(l_m - 1)$ to $\sim 2 \times 10^{-6}$.
- Typical grids ($n_v = 30$, 16+24 polar, 16 azimuthal nodes per channel) give σ_F to $\sim 10^{-4}$ relative.

Inputs and outputs. The user supplies $\sigma_i(W^2, Q^2, t)$ in nb/GeV^2 (class `StructureFunctions`; demo $\rho^0/\phi/J\psi$ models included, to be replaced for physics). `rc_factor(...)` returns $\sigma_{\rm Born}$, $\sigma_{\rm obs}$, $\eta = \sigma_{\rm obs}/\sigma_{\rm Born}$ and all pieces of (12), at fixed $(E_{\rm beam}, Q^2, W^2, t, \phi, h)$ and missing–mass cut. A `TwoSF` wrapper zeroes $\sigma_{TT}, \sigma_{LT}, \sigma_{LT'}$ for direct comparison with the DIFFRAD–like two–SF treatment. Because the SFs are sampled at (17), the model must be defined down to threshold in $\tilde{W}^2$ and for $\tilde{Q}^2$ both above and below the observed Q^2.

6 Pseudoscalar mesons

Nothing in Secs. 2–5 refers to the spin of the produced meson: the five–SF decomposition (10)–(11) is precisely the one introduced for exclusive *pion* electroproduction (it is EXCLURAD's original case), so the identical formulas and code apply to $ep \to e'p\pi^0$, $e'n\pi^+$, $e'p\eta$, with two mass substitutions:

- the detected–meson mass $m_V \to m_h$ ($m_{\pi^0}, m_{\pi^+}, m_\eta$) in the hadron kinematics and in $t_{\rm min}$;
- the *undetected recoil–baryon* mass m_U (proton for $\pi^0 p$, ηp and the vector channels; *neutron* for $\pi^+ n$) everywhere the missing system appears: in the inelasticity $v = \Lambda^2 - m_U^2 = M_X^2 - m_U^2$ and its limits (globally $v \le (W - m_h)^2 - m_U^2$), and in the Li_2 argument of δ_{VR}, Eq. (15), which

becomes $\mathrm{Li}_2[1 - Q^2 m_U^2/(S_0' X_0')]$ (it originates from the soft–photon integral in the rest frame of Λ, whose mass at $v \to 0$ is m_U).

In `vmrad` the SF model simply carries `mV` and `mU` attributes (demo models `demo_pi0`, `demo_pip`, `demo_eta` are provided); the Born closure and IR validation tests pass identically for the pseudoscalar case, including $m_U \neq M$ ($\pi^+ n$). For π^+ production the emission from the charged meson line (and from the hadronic line in general) remains neglected, exactly as in EXCLURAD.

Limitations. Lepton–line radiation only (hadronic radiation and two–photon exchange neglected, as in DIFFRAD/EXCLURAD); $\delta_{\mathrm{vac}}^{\mathrm{had}}$ from a fit; exclusivity is assumed perfect below the missing–mass cut (no detector response). The correction is defined for the unshifted–bin observable $d\sigma/dQ^2 dW^2 dt\, d\phi$ with t reconstructed as $(q - p_v)^2$; if your analysis reconstructs t differently (e.g. from the recoil proton), the shifted variables (17) must be adapted.

References

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